

AIRMAZING

The LSE Generate Global Innovation Challenge

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The Problem | Air Pollution

Every eight minutes, air pollution kills someone in Vietnam.

70,000 deaths a year. One more before this pitch is over.

www.who.int/vietnam

THE CURRENT GAP LIES IN:

01 Vague acknowledgement of personal exposure

02 Root causes of pollution sources

03 The lack of local context

www.cleanairfund.org wvietnamnet.vn/hanoi

Our Solution | *AirMazing*

WHY THIS SYSTEM

01 Personal exposure score

Today's and tomorrow's estimated exposure for a user's commute and schedule, not just a city-wide AQI number.

02 Pollution source attribution

Each source comes with a different recommended action.

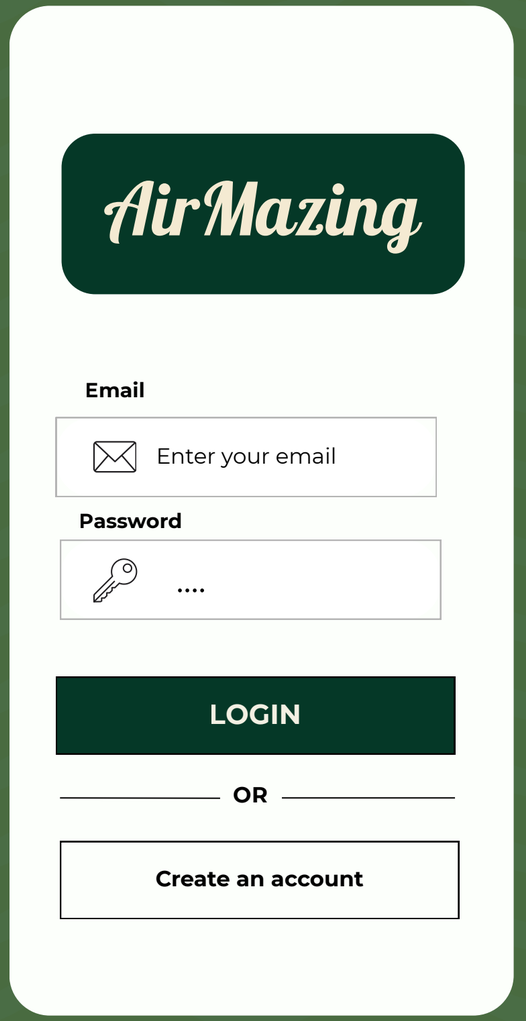
03 Actionable nudge

Leave 30 minutes earlier, take an alternate street, move PE class indoors, or adjust outdoor scheduling for the week.

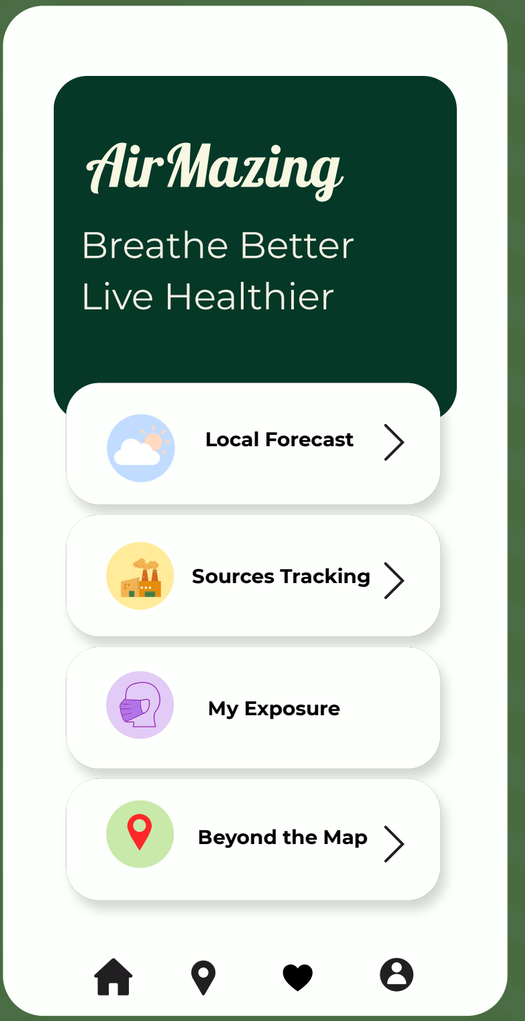
DATA COLLECTION AND INTEGRATION

- Air quality and environment
- Traffic and geospatial
- Source attribution and satellite

APP INTERFACES

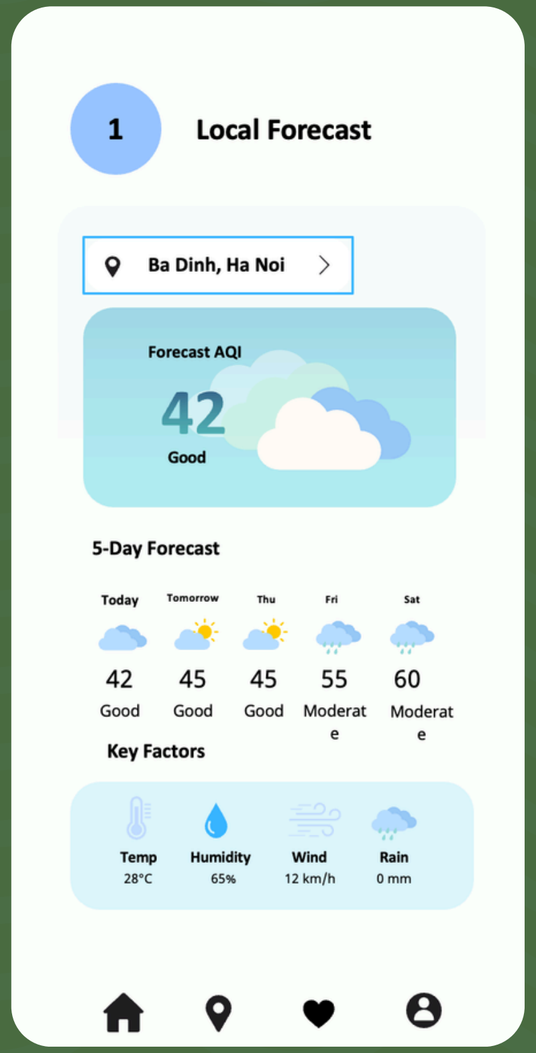


Sign in



Home

Key Features | *Local Forecast*



THE PURPOSE

High-resolution, multi-day air quality forecasts tailored to specific local areas at ward and district level, rather than generalised city-wide averages.

THE APPROACH

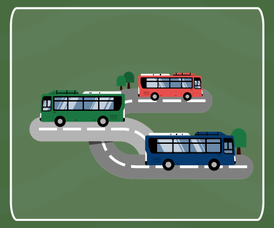
01 Open data, in real time

Historical trends, weather conditions and real-time temporal features from open-data networks (WAQI and OpenAQ).

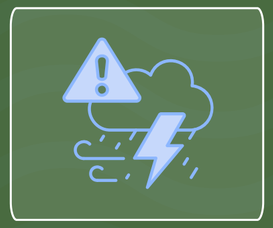
02 Tuned for Vietnam

Vietnam-specific factors such as traffic patterns, crop burning and seasonal weather, used to predict upcoming shifts in pollution.

FACTORS WE MODEL



Traffic patterns



Local Weather

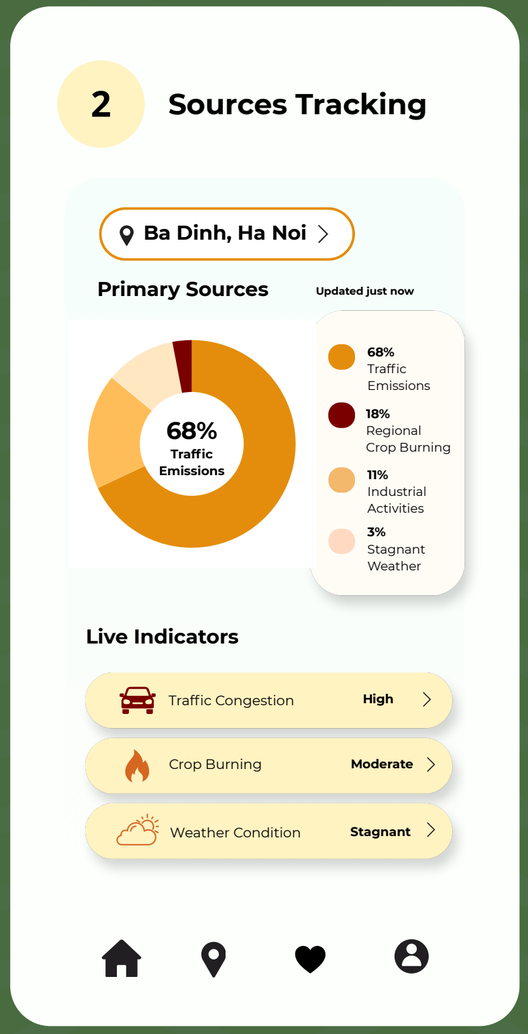


Crop Burning



Industry

Key Feature | *Sources Tracking*



THE PURPOSE

Identify the primary drivers of poor air quality from real-time figures: peak-hour traffic congestion, regional crop burning, or stagnant weather.

THE PROCESS

PHASE 1

Rules first

A rule-based classifier reading wind direction, NASA FIRMS fire hotspots and traffic patterns.

PHASE 2

Label the data

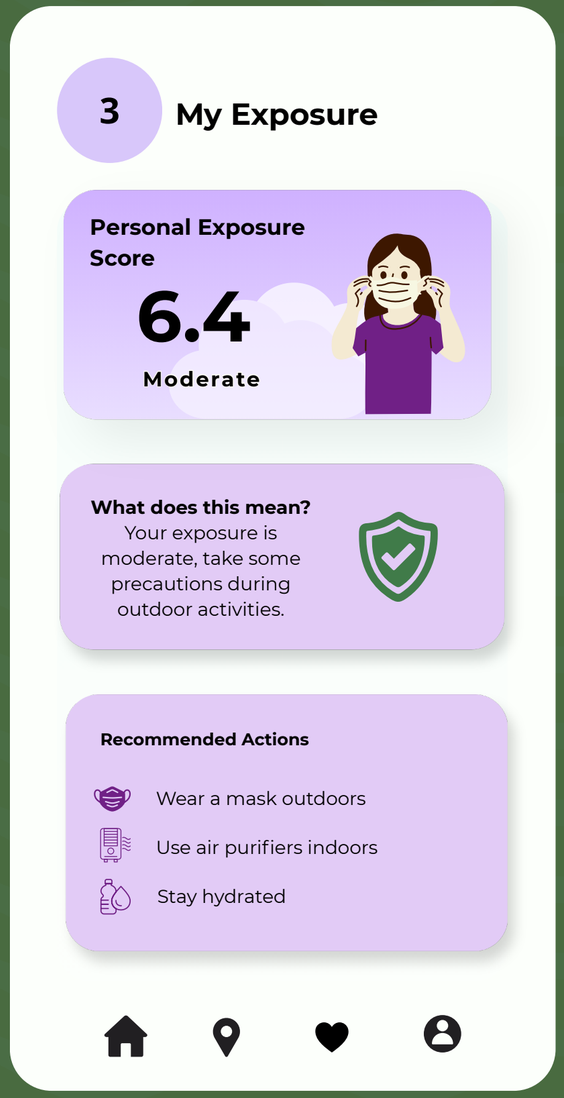
Systematically build a labelled dataset to move toward automated, machine-learning-based attribution.

PHASE 3

Learn and refine

Deploy AI-driven source attribution and keep refining it on real-time data and user feedback.

Key Features | *Personal Exposure*



THE PURPOSE

Translate abstract forecast numbers into an individual's actual exposure level and actionable, real-time health advice.

BASELINE FORMULA

Personal Exposure Score (PES) = AQI (forecast) x M (vehicles modes) x T (outdoor duration)

ELEMENTS

AQI (forecast)

Accounts for varying exposure levels based on commuting habits

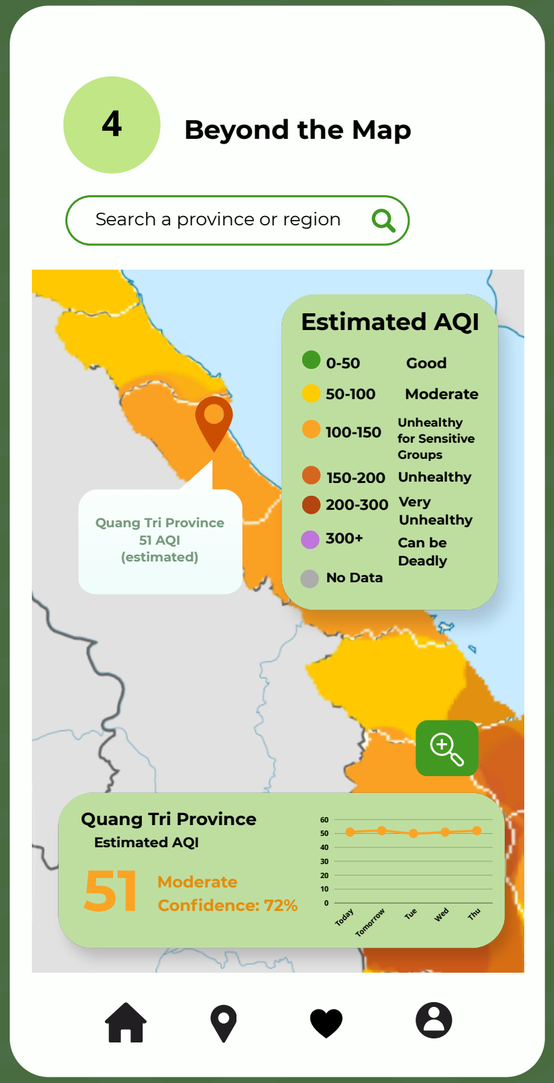
M (Transport Mode)

The proportion of time an individual spends outdoors or commuting daily.

T (Outdoor Duration)

Predicted air quality index for the user's specific location

Key Features | *Beyond The Map*



THE PURPOSE

Estimate air quality for regions and provinces across Vietnam that **lack official monitoring stations** but share **similar geographical and topographical traits** with major hubs like Hanoi or Ho Chi Minh City.

THE APPROACH

Utilize **land-use mapping** (OpenStreetMap) combined with **terrain features, calibrated** against existing baseline ground stations.

Over 34 regions provided, including cities such as:



Hanoi

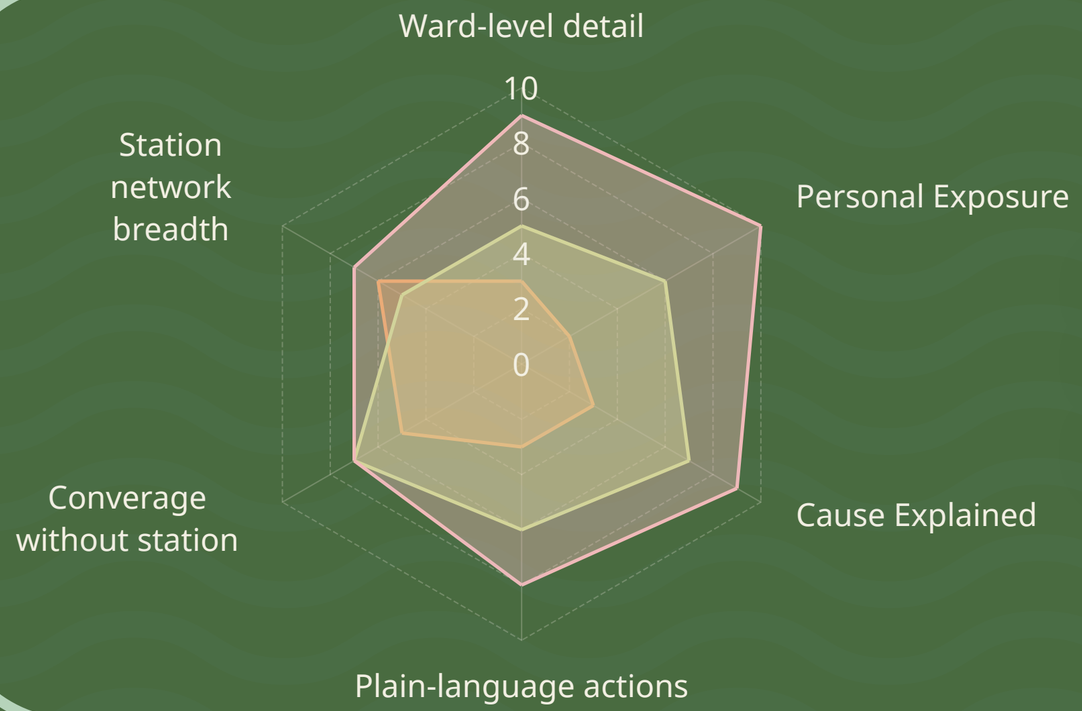


Da Nang



Ho Chi Minh

Competitors Analysis



Existing solutions rarely tell users how pollution affects them personally or what they should do next.

This gap is particularly pressing in densely populated cities, where frequent exposure to poor air quality can become normalized.

Our product focuses on transforming raw data into personalized guidance tailored to each user's health, activities, and surroundings.

Our 4 Advantages

BEYOND AQI

City-level forecasts are commoditized, but we make them explainable with root causes and personalised actions.

PERSONALISED EXPOSURE

We compute actionable exposure levels based on route, location, and outdoor time, not just city-wide readings.

POLLUTION, EXPLAINED

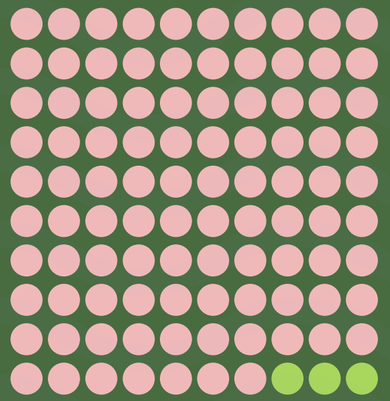
We trace pollution spikes to wind, fire hotspots, and traffic patterns, making the causes clear and actionable.

USER-FRIENDLY

We turn complex data into simple, actionable guidance for students, the elderly, and everyday users.

Scale Of Our Impact

97.1% **of the population lives in air-polluted environment**



Each dot is one percent of the population

70,000+

death caused by air pollution annually in Vietnam.

1.4 millions

years cut from average life expectancy.

	Lower Exposure, Better Health **6-13% reduction** in all-cause mortality for **every 10 µg/m³ reduction** in PM 2.5 concentration	**6-13%**
	Fewer Hospital Visits **28% lower risk** of hospiatl admission for respiratory diseases	**28%**
	Better Learning and Cognitive Performance **5-10% improvement** in cognitive performance	**5-10%**
	Economic Benefits **About 4% of GDP** in Vietnam could **be improved** from meeting WHO guidelines	**$13 billion**

SDG 3: GOOD HEALTH AND WELL-BEING

Personalizes air-quality alerts and guidance to reduce health risks, especially for vulnerable groups.

SDG 11: SUSTAINABLE CITIES AND COMMUNITIES

Supports safer decisions, and promotes healthier, greener, and more resilient communities.

SDG 13: CLIMATE ACTIONS

Raises pollution awareness, encourages adaptive behaviors, and strengthens community resilience.

Timeline

Our business models are **freemium access, premium subscription, B2B partnerships and advertising revenues** Starting in Vietnam, we target a growing global demand for clean-air solutions, driven by industrialization, climate change, and worsening wildfires.

present

1

2026

BUILD AND TESTING

Four core features built into the app. Checked with meteorologists and researchers.

2

2026-27

PILOT LAUNCH

Prototype trials with recruited volunteers in Hanoi and HCMC. First investor pitches.

3

2027

PROVE THE MODEL

Core features first. Revenue validated, costs and funding locked in.

4

2027-28

MARKET ENTRY

iOS and Android. Campaigns, a bigger team, school and business partners.

5

2029+

EXPANSION

Impact investors and NGO funds. ASEAN neighbours, the same air.

As deployments grow, we hope to build a global, distributed air quality network that expands access to reliable air-quality information while delivering highly personalized and accurate insights for every user, regardless of where they live.

Appendix: Predicted Growth

2026

Pilot testing

2 cities (Hanoi + Ho Chi Minh City)
3 pilot schools/NGOs per city
30 users per partner
100+ users total
$0 revenue
$0 profit
100+ users receiving personalized air-quality guidance

2027

Initial commercialization

1,000 free users
30 premium users
10 paying schools
$900 premium revenue
$3,000 school revenue
$50 advertising revenue
$3,950 revenue
$3,500 total cost
$450 profit

2028

Market entry

5,000 free users
150 premium users
30 paying schools
$4,500 premium revenue
$9,000 school revenue
$250 advertising revenue
$13,750 total revenue
$12,000 total cost
$1,750 profit

Appendix: Pilot Testing

Does the app help users reduce their personal PM2.5 exposure?

01 Recruitment



10-20 volunteers wearing a low-cost PM2.5 sensor

02 Randomized Crossover Pilot Trial

Personalized app **VS** Basic AQI

2 weeks each in a randomized order

03 Measurement



Personal PM2.5 exposure level

04 Comparison



Paired statistical test

SAMPLE RESULTS

Basic AQI only

1.32

Exposure ratio

VS

Personalized app

1.12

Exposure ratio

→ **15%**

Lower exposure ratio

8 / 8 participants experience reduced exposure ratio when using the app.